Movies Rating Prediction using Supervised Machine Learning Techniques

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Movies Rating Prediction Using Supervised Machine Learning Techniques

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ABSTRACT
Social media are the most enormous and fast data technology on the internet. A massive amount of data is generated from the internet day by day. The efficient processing of such massive records is hard, so we require a system that learns from these facts and make a useful prediction like machine learning. Machine learning strategies make the systems learn it. We purposed a model for a movie recommendation system using supervised machine learning techniques including K-nearest neighbors (KNN), Support Vector Machine (SVM), and Random Forest Model (RF). In addition to the comparison of all three models on the basis of accuracy, precision, recall, and F1 score, this study also evaluates to see how these three techniques compare against each other. First of all, we discuss a summary of machine learning techniques used for recommendations of items in past. After that, we highlight the most efficient machine learning model for movie rating predictions and recommendations. K-nearest neighbors (KNN), Support Vector Machine (SVM), and Random Forest Models are used here in order to make predictions about movie ratings. The efficacy of all models was tested with Movie Lens 100k and 1M dataset. The random forest Model performs better as compared to other models. Furthermore, we summarize the current challenges based on state of art approaches. Finally, we present open issues and future directions for further research.

Keywords: Random Forest Model, K-Nearest Neighbor, Support Vector Machine, Recommendation System Random Forest (RF) Model, K-Nearest Neighbor (KNN), Support Vector Machine(SVM), Recommendation System (RS)

1. INTRODUCTION

Recommendation systems (RS) help users by suggesting items according to their preferences and interests. Recommendation systems (RS) is software that provides personalized recommendations when there are many possible alternatives available as given in figure 1.

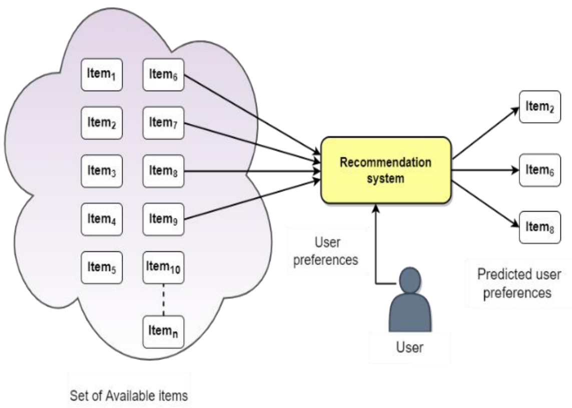

Figure 1. Recommendation system

The recommender system recommends items to the users according their preferences. Recommender system take verities of multiple items as an input and filter out relevant items and predict the user preferences. A recommender system is a filtering system that makes an accurate prediction regarding the "rating" or "preference" a person would assign to a given item. It's not uncommon to use a recommendation system. Examples include social networking platform content recommendation engines, product recommendation engines for online retailers and play list generators of video and audio streaming services. By analyzing a user's preferences and interests, a recommendation system (RS) may exclude any irrelevant information or items and would instead provide suggestions directly related to the user's given preferences. Recommendation systems (RS) prediction and rating based on user interests, browsing history, and preferences. Several web-based services are available on the internet, day by day

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there is an addition of thousands of products and services on E-commerce sites, as the result, these sites face the information overload problem. To overcome this problem many researchers proposed various methods and techniques [1]. In this study the author employed the Collaborative filtering recommendation technique in E-Learning with most of the studies about improving the standard of recommendations [2]. The Author proposed a machine learning-based algorithm that suggests an eligible applicant's CV to HR based on the particular job description. This approach has an accuracy of 78.53% with the Linear SVM classifier.

Table 1:Popular sites using recommender systems

Sites	What is recommended
Amazon	Books/other products
Facebook	Friends. Pages, groups
Netflix	Movies, Drama, Session
Twitter	Posts, Tweets
YouTube	Videos, channels
Spotify	Music, Audio
Yelp	Food, Restaurants
LinkedIn	Job, Connections
Trip Advisor	Hotel
TED	Articles

Table 1 lists some well-known websites that use recommendation systems for various purposes [3]. The Amazon platform recommend similar products to customer according to their interest and preferences. The Facebook suggest friends, pages and groups according to user behavior, search history and interest. Netflix recommend movies, dramas, documentaries to user similar to user taste and their previous watched history and search history. YouTube recommend videos and channels. Twitter recommends similar tweets and posts; Spotify recommend music and audio songs to their audience. Yelp recommend food and restaurants to their customers. LinkedIn recommend jobs and connection related to user profile. Trip Advisor suggest hotel and TED suggest articles according to user preferences. Many companies grow their business and revenues by using recommendations engines [4]. In order to save consumers time, movie recommendation systems suggest films for them to watch rather than requiring them to go through collections of movies numbering in the thousands or even millions. To save time and effort, we will provide movie recommendations based on the user's preferences [5].The recommendation system also reduces the time complexity issue by suggesting user the valuable contents according to user preferences and interest. Now a days many platforms make use of recommendation engine to grow their business such as Amazon, Netflix, YouTube, LinkedIn and Spotify. In this work, we employ collaborative filtering to make predictions about users' movie ratings based on their age, gender, profession and previous movie preferences. We use publicly available dataset from movie lens database to test the efficiency of models. Hence it has been proposed utilized three supervised machine learning models including KNN, SVM, and Random Forest to predict movies ratings. Furthermore, all classifiers have been compared against each other in terms of accuracy. The random forest model achieves high accuracy as compared to others.

The main objective of this study are to highlight the most efficient and accurate supervised machine learning model for movies rating predictions and recommendations. This paper presents the studies related to recommendation systems. One of our primary goals is to find the most effective supervised machine learning method for predicting and recommending movie rating. We implement different classification models and evaluate their performance in term of accuracy. We perform Data Preprocessing and perform Implementations of three supervised machine learning classifiers to predict movies ratings. To address the sparsity problem in the recommendation system, we designed three distinct models and compared their results using several metrics. The best model was determined using a matrix that included accuracy precision, recall and f1-score. The Implementation of all models done using python language in googles Collab. Labeled datasets are used in supervised machine learning to train algorithms how to classify data or make accurate predictions. Supervised Machine Learning allows you to use input label data and produce output from the preceding experience. When applied to real-world computing problems, it may be quite effective. While a wide range of machine learning techniques has been implemented to the task of making suggestions in the past [6], we concentrate here on the specific subset of these techniques known as supervised machine learning algorithms, that are employed to generate recommendations and to evaluate competitive approaches to find the most effective one.

The remaining portions of this article are arranged as following. The section 2 begins with a description of the available literature on the Movie Recommendation System also discusses Existing work about the movies recommendation systems. A literature table of Supervised Machine Learning techniques, which may be utilized for movie recommendation systems, is presented. At the end the table represent application of Recommendation system. Section 3 is comprehensive explanation of the proposed Predicting and suggesting films based on their ratings. Chapter 4 describes the system's layout and implementation. Section 5 will include the experimental work and findings, while Section 6 will present a summary of the study and recommendations for further research.

2. LITERATURE REVIEW

Machine learning makes use of Content-based Filtering to identify relationships between features based on their common attributes. Recommender systems, which are designed to suggest customers' products according to their interests and preferences, used the filtering strategy. Content-based filtering techniques (CBF) filter the things liked by the user and also recommend similar things to same user. A machine learning technique called content-based filtering creates choices based on similarities between characteristics. Content-based filtering doesn't need the data of other users to provide suggestions to one individual [7]. The CBF algorithm suggests products that are similar to ones that were previously liked. It looks at previously rated things and suggests the best match [8]. It is further stated that CBF, is based on content correlations. To determine the similarities between objects, CBF employs item information, which is expressed as attributes. A machine learning method called content-based filtering bases choices on the similarity of features. This tactic is widely used in recommender systems, which are algorithms that provide product recommendations to consumers based on data obtained about them [9]. This article uses a content-based movie recommendation system that correlates genres to provide smart suggestions to users [5]. For each user, content-based filtering looks at their past behavior and providing suggestions based on their common features as mention in Table 2. By comparing films' genres, this aims to suggest movies to viewers. If a user has given a certain film a high rating, the algorithm will suggest further films in the same genre.

Table 2. Application using Recommendation System

Applications	Recommendations	Benefits
YouTube	Recombed videos and channels	YouTube use deep neural networks for recommending videos. Assisting people in finding the videos they want to watch Locating previously watched videos with ease Recommend channel for you Recommend Popular Music Videos Recommend Trending Sports Videos Recommend Popular Gaming Videos Recommend Popular Films Videos
Facebook	Recommend New friends, pages and groups	Recommend friends for the Facebook users Recommend pages according to user interest Recommend relevant groups
Instagram	Hashtag Suggestions suggests posts	Every month, they can explore new photographs, videos, and Stories that are relevant to their interests. Users can get Instagram's value-added information (or photos) via the recommendation platform.
LinkedIn	Recommend New connections, profiles	LinkedIn Job Recommendations displays job opportunities based on your job searches, job alerts, LinkedIn profile, and activity.

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Amazon	Recommend similar products	Amazon's recommender systems are able to automatically analyze and predict a customer's buying preferences, then expose them to a choice of products that might attract to them. Amazon uses item based collaborative filtering, which generates high-quality real-time recommendations.
Netflix	Recommend Movies	Recommend movies to its customers based on their viewing history and user ratings

Recommendation system have used a technique called Collaborative Filtering (CF). Collaborative filtering is an approach to user interest prediction that makes use of shared characteristics between users and the contents they watch. CF techniques based on item to item and user to user similarities. Things that are liked by one user also recommend to other user that have similar interest and preferences [10]. The two most commonly utilized categories of CF methods are memory-based methods and model-based approaches. For the memory-based system to function, all ratings, objects, and users must be stored in memory. Model-based methods, on the other hand, frequently produce a summary of rating patterns [11]. Item-based CF techniques offer recommendations that are simple to implement, efficient, and justifiable, because of their effectiveness, collaborative filtering techniques are quite popular. These CF methods are capable of producing standard recommendations for a wide range of problems [12]. Moreover, use of CF, a user may obtain a suggestion for in an item based on ratings and reviews submitted by other users since the system looks at past user behavior to establish a link between individuals and goods [13]. Memory and model-based techniques are used in collaborative filtering. To provide suggestions, memory-based systems examine data provided by the user. Incorporating user evaluations promotes their accuracy. Model-based techniques create a profile of a user's previous actions that help anticipate their future behavior. Accuracy and scalability are greatly enhanced by the use of partitioning-based approaches.

Hybrid methods constitute a type of recommendation system which incorporate elements of both CF as well as CBF. The advantages of both collaborative recommendation systems and content-based filtering are brought together in the Hybrid Approach. Recommender systems are software systems that use various methodologies to produce and deliver suggestions for items and other things to users [14]. The hybrid recommender system construction technique presented in this study combines the Self-Organizing Map neural network technology, Collaborative Filtering, and Content Based Approach [15]. Despite their widespread use, these filtering techniques suffer from many drawbacks that lead to unsatisfactory suggestions. Examples include sparsity, scalability and Cold Start Problem. Hybrid filtering is superior to both content-based filtering and collaborative filtering [13]. To improve recommender systems and remove the limitations from each, hybrid systems integrate social and content-based filtering techniques.

Two online datasets Movie Lens and Netflix are mostly used for movie ratings. Despite being one of the techniques for predicting movie ratings that is most often used, this algorithm appears to have a scalability issue. Table 3 provide a detail of literature review with respect to dataset, algorithms and their accuracy.

Table 3. Literature Review

Ref.	Year	Techniques/Algorithm	Dataset	Accuracy	Future Work
[16]	2020 Journal paper	Collaborative Filtering (CF)/ algorithm Weighted KM-Slope-VU	Movie Lens data	reduce time complexity.	Use fuzzy C-means to further improve the accuracy of recommendations
[17]	2019 Journal paper	Matrix Factorization (MF) and Recurrent neural network (RNN)	Netflix dataset and Movie Lens dataset	improve performance	Add more data elements such as movie details, movie reviews, and user profile details to further improve performance.
[18]	2020	hybrid recommendation system (RS)	Twitter tweets	Compared to the other models, the	Consider more details from various social media platforms

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	Journal paper	Weighted Score Fusion	movie metadata IMDb Movie Ratings	suggested model provides more accurate recommendations.	regarding the user's emotional tone.
[19]	2019 Journal paper	HI2Rec method	MovieLens1M datasets	Improve the effectiveness of recommendations by resolving issues with data scarcity and a cold initialization.	Apply it to additional fields like music, e-commerce, news.
[20]	2019 Journal paper	Aesthetic aware Unified Visual content Matrix Factorization (UVMFAES) Method	Movie Lens and IMDb2 datasets.	recommendations more accurate	To improve video characterization, multimodal fusion is being considered.
[21]	2020 Journal paper	MFMP Model (Markovian factorization of matrix process) a context-based recommender system (CRecSys)	MovieLens10M and MicroVideo-1.7M	By a significant margin, the proposed method outperforms existing state-of-the-arts.	To better characterize videos, we intend to use multimodal fusion.
[22]	2020 Journal paper	Markovian factorization of matrix process (MFMP)	Movie Lens dataset	superior model to time SVD++ and a standard tensor factorization model	Add some context to the rating to solve the "cold start" or "early voter" problems.
[23]	2020 Journal paper	linear mixed model LINEAR MIXED-EFFECTS MODEL (LMM)	Movie Lens dataset	Deal with cold start problem	Combine a model like this with the traditional SVD approach. Utilize a multi-source data set.
[24]	2020 Journal paper	context-based recommender system, CRecSys	MovieLens-1M dataset IMDB dataset	Perform better handle new user and limited content issues	Using deep learning techniques to further improve performance.
[25]	2019 Journal paper	multi-criteria recommender systems (MCRSs)	users' reviews and ratings	Achieve better accuracy and Deal with two issues cold start and data sparsity issues	investigate the reviews' implicit values
[26]	2020 Conference paper	K-NN algorithms Cosine similarity Item-based collaborative filtering	Movie lens dataset	enhance accuracy and performance	Datasets may be expanded to contain more information (e.g., release date, actors' names, genres and roles).
[27]	2020 Journal paper	f multilingual twitter sentiment analysis (MLTSA) Random Forest (RF), Logistic Regression (LR), Support Vector Machines (SVM), Decision Tree (DT), Multinomial Naïve Bayes (MNB), k-Nearest Neighbor (KNN).	Dataset Twitter API	RF (random forest), it is 93% SVM (support vector machine) is 95%	When words as well as sentences are code-mixed and switched, accuracy reduce.
[28]	2019	Random Forest	Amazon dataset from	Random Forest 95.03 %	improving results by employing a hybrid approach or lexicon-based approach.

	Conference paper		www.kaggel.com		Use more than one language
[29]	2020 Conference paper	SVM KNN Sentiment analysis of Movie reviews text mining the NLTK (Natural Language Toolkit) module	Data is obtained from the TMDB API	SVM 85%	enlarge our rating system to include a movie recommendation feature
[30]	2019 Conference paper	K-Means Clustering, KNN, Collaborative Filtering	Kaggle website	Similarly, reducing the total number of clusters leads to a lower RMSE value. The ideal RMSE was 1.081648 obtain.	Sentimental Analysis concept can be used in future to enhance the efficiency
[31]	2019 Conference paper	Naive Bayes Item-CF, and User-CF	Movie Lens data set	The Naive Bayes algorithm performs better.	Increase the number of users in the recommendation to further boost efficiency
[32]	2018 Conference paper	Naive Bayes (NB) as well as Support Vector Machines (SVM)	Bangla Movie Database (BMDb)	SVM achieved to some extent better results with precision of 0.86.	Semantic analysis for massive data sets is essential in this area to study more effective linguistic expertise enabling sentiment extraction.
[33]	2020 Journal paper	K-Nearest neighbors (KNN), Support-Vector Machine (SVM), Logistic Regression (LR), Multinomial NB(MNB) and Multi-layer Perception (MLP)	200k Headlines of News	Accuracy of the Support-Vector Machine approximately 75.13 %.	Utilize different word or embedding sentence victimizers to enhance the model
[34]	2020 Conference paper	Cosine Similarity and KNN.	use dataset of movies with different genres	accuracy improved	Advanced deep learning techniques used in future to further improve accuracy.
[35]	2020 Conference paper	Multilingual twitter sentiment analysis (MLTSA) with SVM	Twitter	reduce data sparsity issue SVM 95 %	When tweets use code-mixed and code-switched words and sentences, accuracy suffers.
[36]	2017 Conference paper	Maximum Entropy	Twitter	Maximum Entropy 74 %	The main difficulty is in analyzing the tweet dataset in multiple languages.
[37]	2019 Conference paper	Naïve Bayes classifier.	IMDb (Internet Movie Database) web portal at http://reviews.imdb.com/Reviews	Naïve Bayes classifier. 81.45 %	To improve accuracy, a hybrid technique that applies the permutations and combinations of the aforementioned classifiers can be used.

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[38]	2020 Conference paper	RNN	Twitter API	RNN 91.67 %	In preprocessing, emotions added.
[39]	2018 Conference paper	Naive Bayesian classifier with Gaussian correction and feature engineering	Movie lens 100k data set	improve performance	Newly developed features that are related to user data produce better results and raise the standard of recommendations.
[40]	2019 Conference paper	Doc2vec model	Movie Lens dataset movies.csv, ratings.csv.	To improve accuracy and Recall	In future deep learning model used to achieve better accuracy.
[41]	2018 Research Articles	hybrid Approach	movie dataset (https://movie.douban.com/).	Enhance the recommendation system's timeliness and accuracy	concentrate on removing from users any personal characteristics that may be concealed in the text description
[42]	2020	Generalized Linear Model (GLM)	Movie Database (IMDB)	Due to its lower value of RMSE, GLM is the high-achieving accuracy regression classifier.	future research on the popularity of the movie using advance techniques.
[43]	2019 Journal paper	Naïve Bayes algorithm	Data from Movie Database (IMDB) containing 1000 positive reviews and 1000 negative reviews	Naïve Bayes algorithm outperforms	Boost the real-time accuracy of movie recommendations.
[44]	2020	Content Based Filtering, Collaborative Filtering, hybrid filtering Deep Learning Based Methods	movies dataset	Building an effective recommendation system with hybrid filtering	Any recommender system's accuracy can be increased by including additional movie features.
[45]	2017 Conference paper	Convolution Neural Network and Bi-directional LSTM RNN.	IMDB dataset contains 25000 movie reviews	improves accuracy compared to CNN, LSTM and Bi-LSTM model	Attempt to discover a deep learning technique that works much better for analyzing emotions. Acknowledgments

[46]	2018 Journal paper	Propose CAM based on Convolutional neural network (CNN) Model	IMDB (English) and WATCHA (Korean) datasets.	classify the sentence's polarity correctly Identify words with a strong polarity.	Another interesting area for future research could be the creation of a structured and quantitative evaluation method for word localization.

3. METHODOLOGY

K-Nearest Neighbors (KNN), Support Vector Machine (SVM) and Random Forest Model (RF) models were utilized for supervised machine learning in this paper for a movies rating prediction and recommendations and also give comparison of all models in term of accuracy which model perform better for rating predictions based on users' ratings and demographic features such as movie id, user id, age, gender, occupation as the result the data sparsity issue is resolve and better accuracy achieve. Figure 2 show the flow of Research Methodology adopted to conduct the study, Figure 3 represent the dataset.

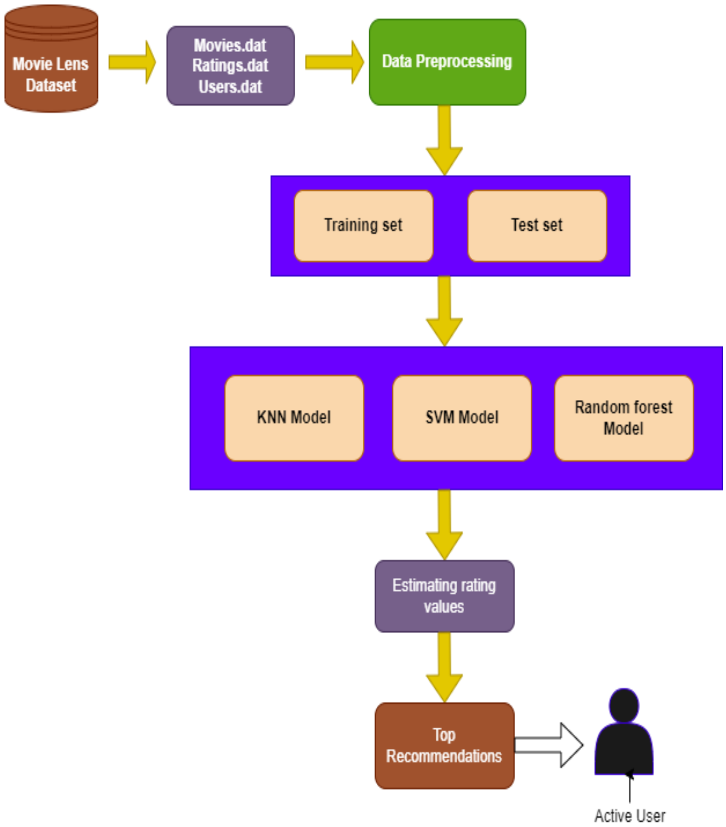

Figure 2. Methodology adopted to conduct the study

The datasets are taken from the Movie lens site (https://grouplens.org/datasets/movielens/). Three file Movie's data, rating data, and user data are used in this paper. The training set comprises 80 percent of the dataset, with the remaining 20 percent serving as the test set. Data from the Movie Lens 100k and 1M datasets were used to create the model's implementation. In movies dataset column 0 represent movie ID, column 1 represent the name of movies and columns 2 represent the types of movies or categories of movies as given in the Table 4. Movies dataset contains 3 columns and 3883 rows and records. Rating dataset contains 4 columns and 1000209 rows. column 0 represent movie ID, Column 1 contain user ID, column 2 represent movie ID, column 3 represent Rating of users range from 0 to 5 the maximum rating value is 5 and the last column represents the timestamps information. There are 6040 rows and 5 columns in the user dataset. Zip code, gender, age, occupation, and user ID

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are all recorded in columns 0, 1, 2, 3, 4 respectively. In this study, we predict and recommend movie ratings based on the user's age, gender and occupation.

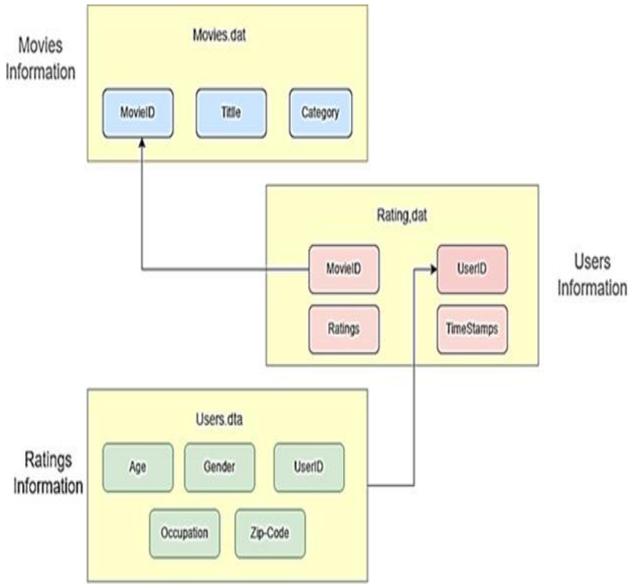

Figure 3. Dataset Representation

Table 4. Dataset Description

Name	Type	Size	Value
movies	DataFrame	(3883, 3)	Column names: 0, 1, 2
Rating	DataFrame	(1000209, 4)	Column names: 0, 1, 2, 3
users	DataFrame	(6040, 5)	Column names: 0, 1, 2, 3, 4

The selection criteria for selecting three supervised machine learning algorithm KNN, SVM and Random Forest model is based on the previous studies gabs and literature review. The related studies indicate that the supervised machine learning techniques used in movies rating predictions and recommendations. In past studies mostly the KNN, SVM and random forest models are used to predict movies rating and recommendation with better accuracy. From the related studies we conclude that the KNN, SVM and Random Forest models outperform have greater accuracy as compared to other techniques. We consider three models and compare their performance in term of accuracy. Supervised machine learning model, such as KNN. We train the model by using labeled data as input. The classification tasks are solved using this model. This algorithm is called K-Nearest Neighbors Algorithm. As a supervised machine learning method, it requires labeled data to function and generates results when provided with new, unlabeled data. In this Model, the quality of the predictions depends on the distance measure between the two nearest points. In this paper, K-Nearest Neighbors (KNN) is an algorithm used in machine learning to categorize individuals into categories having common interests in movies. The KNN algorithm works on the hypothesis that items with similar features are relatively connected to each other.

The supervised machine learning method known as support vector machine (SVM) can resolve nonlinear as well as linear classification and regression challenges. Support Vector Machines (SVM) are fast and reliable classification algorithm that performs well with small amounts of data. The objective of Support Vector Machines (SVM) is to identify a maximum marginal hyperplane by classifying datasets (MMH). Those points of data set that are in the relatively close proximity to the hyperplane called support vectors. Whenever interacting with a classification or regression issue, supervised machine learning methods are frequently applied. A more advanced variant of the decision tree known as Random Forest uses numerous classifiers rather than just one to predict future events with accuracy and correctness. The Random Forest technique succeeds in both regression and classification because it can process data sets between both categorical and continuous variables. It produces superior results when it comes to classification issues.

4. DESIGN AND IMPLEMENTATION

As either a means of predicting and recommending movie ratings, we use supervised machine learning techniques such as support vector machines, K-nearest neighbor and random forests. In this chapter, we explain how to design and implement SVM, KNN, and Random Forest Classification models. The main parts are Model Building, Training, and Testing and Predictions. Finally, we detail how to evaluate the effectiveness of these models using a variety of performance matrices, including the accuracy score precision, recall and f1score.

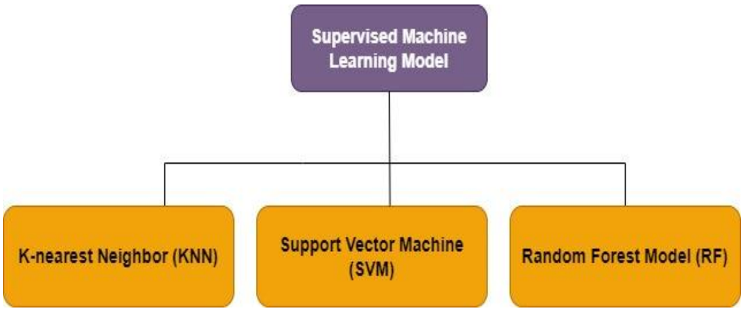

Figure 4. Supervised machine learning models

The purpose of this study is to predict movie ratings using 3 supervised machine learning models (see Figure 4), and Figure 5 represent the implementation of ML models.

Figure 5. Steps in Machine learning model's implementation

a) Data acquisition of the movie lens dataset

The datasets are taken from the open-source Movie lens. Data from the Movie Lens 100k and 1M datasets were used to create the model's implementation. Three Files movie.dat, user.dat and rating.dat are used mentioned in the Table1-7.

b) Data preprocessing

Remove unnecessary columns.

Remove Null or Nan values, Balance the unbalance dataset, handling missing values.

c) Coding Environment

All models were implemented in google Collab environment using Python programming language.

d) Importing Necessary Libraries

The essential libraries like "pandas" "NumPy" are imported into the project.

e) Loading and Formatting the Data

The user.dat, rating.dat and movie.dat are loaded into the project. Unnecessary columns are deleted from the dataset.

f) Perform the Exploratory Data Analysis (EDA) for the user's dataset

Perform machine learning on required extracted records

Create train and test data set. Apply KNN, SVM and Random Forest Models

g) Training of Models

The training set comprises 80 percent of the dataset, with the remaining 20 percent serving as the test set. In this stage the training of models is done by giving the training dataset. Set batch size and epoch value.

Calculate Predictions Accuracy score for KNN, SVM and RF Model

h) Compare Accuracy score of all Models

Performance Evaluation Matrices

Confusion matrices represent counts from predicted and actual values. The comparison of all three models done on the basis of accuracy, precision, recall, and F1 score. Accuracy is a model evaluating matric that is used to check the performance of classifiers. Precision refers to how close or far apart the measurements are. Precision can be seen as a measure of quality. The recall is defined as how many of the true positives were recalled (found). The F1 score measures a performance of the model on a given dataset. The accuracy of a test may be measured by its F1 score. All matrices' formulas given below in Table 5.

Table 5: Performance Matrices

Performance Matrices	**Equation**
Accuracy	$Accuracy = \frac{(TP+TN)}{(TP+FP+TN+FN)}$
Precision	$Precision = \frac{True\ Positives}{(True\ Positives + False\ Positives)}$
Recall	$Recall = \frac{True\ Positives}{(True\ Positives + False\ Negative)}$
F1-score	$F1 = 2.\frac{Precision\ .recall}{Precision + recall} = \frac{TP}{TP + \frac{1}{2}(FP+FN)}$

5. RESULTS AND DISCUSSION

This section comprises of discussion on the results obtained from the study.

Table 6. Analysis of Movies dataset

	MovieId	**Title**	**Genres**
0	1	Toy Story (1995)	Animation Children's Comedy
1	2	Jumanji (1995)	Adventure Children's Fantasy
2	3	Grumpier Old Men (1995)	Comedy Romance
3	4	Waiting to Exhale (1995)	Comedy Drama
4	5	Father of the Bride Part II (1995)	Comedy
5	6	Heat (1995)	Action Crime Thriller
6	7	Sabrina (1995)	Comedy Romance
7	8	Tom and Huck (1995)	Adventure Children's
8	9	Sudden Death (1995)	Action
9	10	GoldenEye (1995)	Action Adventure Thriller

Table 6 presents the movies dataset contains three columns MovieID, Title and Genres of movies.

Table 7. Analysis of Ratings dataset

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UserID	MovieID	Rating	Timesstamp
1	1193	5	978300760
1	661	3	978302109
1	914	3	978301968
1	3408	4	978300275
1	2355	5	978824291

Table 7 presents the rating dataset contains four columns userID, movieID, ratings and timestamp values. The rating values in range between 1 to 5.

Table 8. Analysis of Users dataset

UserID	Gender	Age	Occupation	Zip-Code
1	F	20	10	48067
2	M	56	16	70072
3	M	25	15	55117
4	M	45	7	02460
5	M	25	20	55455

Table 8 illustrates that the user dataset includes relevant data such as User ID, gender, age, occupation and zip code for each subscriber.

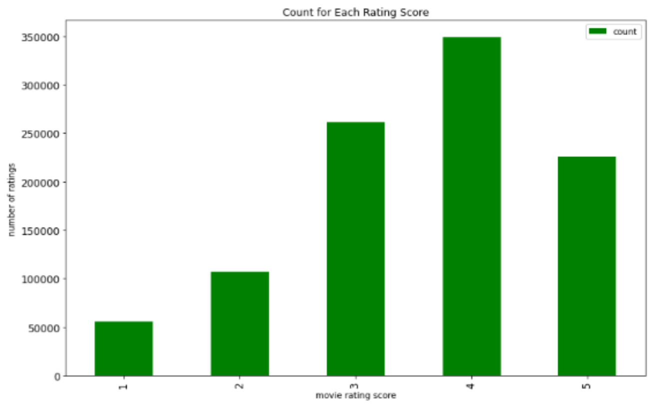

Figure 6. Bar chart of rating score count

The Figure 6 shows the bar chart of movie ratings. The user rate movie in range 1 to 5. From above chart we observe that the most rated rating value is 4 than 3, 5, 2 and 1 rating value given to the watched movies. In dataset 350000 ratings have 4 values, 270000 have 3 rating value, 220000 rating have value 3 and 100000 rating have 2 value and 50000 rating have 1 value.

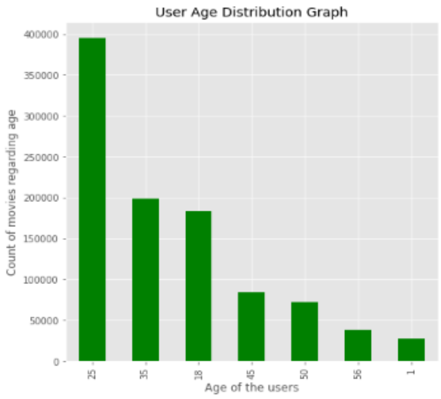

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Figure 7. Bar chart of Movies count regarding Age of the users

The Figure 7 shows the count of movies according to age we observe that the greatest number of movies watched user with 25 age group. 25 age group watched almost 390000 movies.

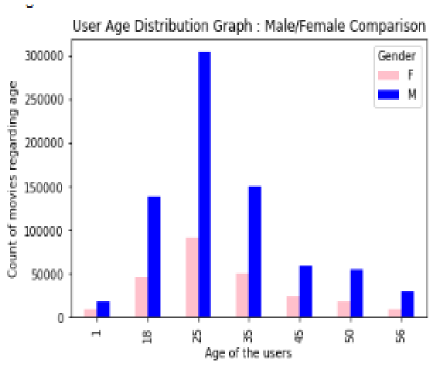

Figure 8. Bar chart of Movies count according to Gender

The Figure 8 shows the count of movies according to gender. The male watched most of the movies and give ratings as compared to female. We observe that the male watched more movies as compared to females and male give more ratings to more movies.

Table 9. Number of movies with rating values

	Count
0	21384031
1	56174
2	107557
3	261197
4	348971

The rating values are 1 to 5, the above Table 9 shows that total count of movies and their corresponding rating vale from 1 to 5 including movies having zero rating value. Dataset set split into 80:20 ration 80% of training data and 20% of test data. We Use the following features: movie id, age, occupation and gender in this work. The datasets were divided into a training data (80%) and test data (20%) set (80:20 ratio). We Use the following features: movie id, age, occupation and gender. We implement three models KNN. SVM and Random Forest model for movies rating predictions. We train all supervised machine learning classifiers on training dataset. The ratio of training and test data is 80:20. 80% dataset used as training set to train the ml models and 20% is used as teste set to validate the models. To achieve this, we use metrics including accuracy, precision, recall and f1-score to examine how effectively each model performs. In this study, we use three supervised machine learning models to make these recommendations. All models are tested and ranked based on their performance using metrics including the accuracy score, classification report and confusion matrix. The weighted average values of Accuracy, precision, recall and f1-score for all models are mentioned below in Table 10.

Table 10. All Models Performance

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
KNN	69	76	69	65
SVM	54	55	54	51
RF	92	92	92	92

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It has been observed that after a careful comparison of all models, the random forest model performs best to with 92 % accuracy to predicting movies ratings as given in Figure 9.

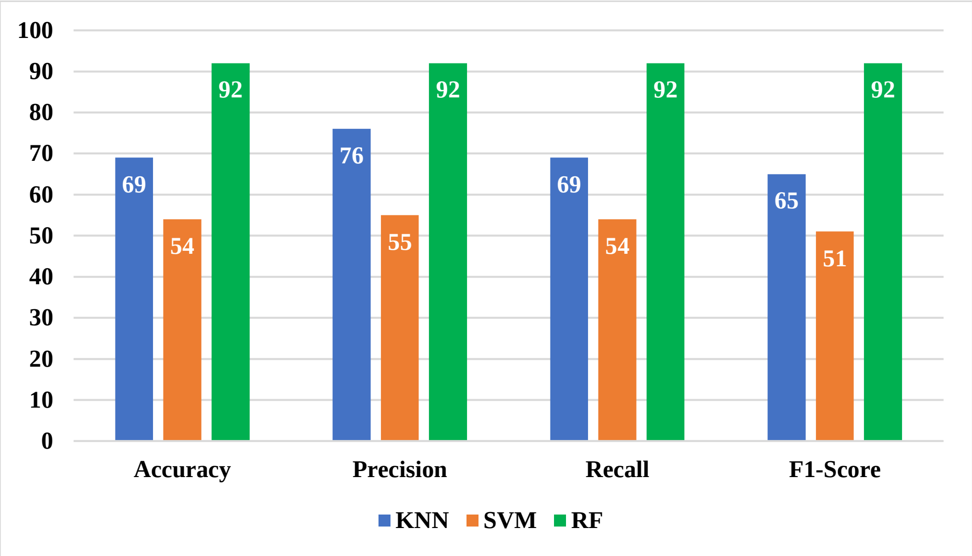

Figure 9. Accuracy, precision, recall and F1-score of KNN, SVM and RF Models

According to the KNN model classification report, the model's accuracy is 69%, with weighted average value of 76% for precision, 69% for recall and 75% for f1-score. SVM model classification report shows 54% accuracy, 55% precision, 54% recall and 51% f1-score. Based on the results of the Random Forest Model's classification study, we find that the model seems to have an accuracy of 92%, with 92% precision, 92% recall and 92% f1-score on average. If we make comparisons of all models, we observe that the random forest model performs best to with 92 % Accuracy to predicting movies ratings as shown in Table 11.

Table 11. The Accuracy Score of All Models

No.	Model	Score
1	Random Forest	92
2	K-Nearest neighbor	69
3	Support Vector Machines	54

The random forest model outperforms as compared to SVM and KNN model.

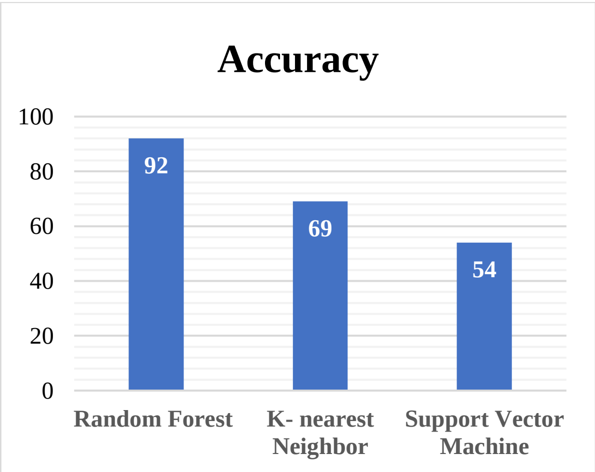

Figure 10. KNN, SVM and RF Models Performance.

The above Figure 10 shows the accuracy score of random forest model is 92, KNN has accuracy score 69 and SVM has 54. So, from the result we analyze that the random forest model outperforms for movies rating predictions as compared to SVM and KNN model in term of its accuracy score.

6. CONCLUSION AND FUTURE WORK

This paper is basically focusing on Movies Rating Predictions and Recommendations. We use movie lens dataset. The efficacy of all models was tested with Movie Lens 100k and 1M dataset. KNN, SVM, and Random Forest are three supervised machine learning models that have been used to predict movie ratings. The primary goal of comparing these three models is to determine which one provides the most accurate predictions of movie ratings using Supervised machine learning. We perform preprocessing of dataset to achieve valuable results. Finally, the experimental study revealed that KNN has a 69% accuracy score, SVM has a 54% accuracy score and the Random Forest model has a 92% accuracy value, indicating that Random Forest model achieves a higher accuracy score as compared to KNN and SVM Models.

It has been planned to upgrade the model in the near future by adding more features in movies dataset for better rating predictions and recommendations. Utilizing movie poster pictures and other visual contents, enhanced movie recommendations are made using Deep Learning methods including Artificial Neural Networks (ANN), Deep Neural Networks (DNN), and recurrent Neural Networks (RNN).

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